OpenEdge Bridge — Documentation

Sample INSERT Data

Note: The sample INSERTs below show the full schema with **one RTC row followed by one OSCA/senior row**, so you can see which columns each kind populates. RTC fills `batch-no`/`app-no`; OSCA/senior fills `control-no`/`id-no` and leaves `batch-no`/`app-no` NULL. `trans-no` is present on both.

MySQL — `id_print` (example: one RTC + one OSCA/senior record)

```
INSERT INTO id_print (trans_no, trans_date, trans_time, batch_no, batch_date, app_no, dob, cp_no, fullname, hcp_fullname, address, hcp_cpno, lname, fname, mname, spa_docno, page_no, book_no, series_no, spa_atty, spa_dt_issue, printed, printed_dt, printed_tm, tran_type, image_data, id_code, sfx, photo, gender, date_issued, id_no, year_applied, control_no, id_type, pvc_no, age, print_count) VALUES -- RTC record (tran_type 1): keyed by batch_no + app_no (1001, '2026-01-15', '10:30:00', 1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr. Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 'no', NULL, NULL, 1, NULL, 5001, '', NULL, 1, NULL, NULL, 0, 0, 1, 'PVC-000001', 0, 0), -- OSCA/senior record (tran_type 2): keyed by control_no + id_no; batch_no/app_no are NULL (2001, '2026-01-16', '09:15:00', NULL, NULL, NULL, '1941-11-18', '', 'Maria Clara Santos', '', '456 Mabini St, Quezon City', '', 'Santos', 'Maria', 'Clara', '', '', '', '', '', NULL, 'no', NULL, NULL, 2, NULL, NULL, '', NULL, 2, '2026-01-16', 'OSCA-2026-0001', 2026, 9001, 1, 'PVC-000901', 84, 0);
```

OpenEdge (Progress) — `pub."id-print"` (example: one RTC + one OSCA/senior record)

```
INSERT INTO pub."id-print" ("trans-no", "trans-date", "trans-time", "batch-no", "batch-date", "app-no", "dob", "cp-no", "fullname", "hcp-fullname", "address", "hcp-cpno", "lname", "fname", "mname", "spa-docno", "page-no", "book-no", "series-no", "spa-atty", "spa-dt-issue", "printed", "printed-dt", "printed-tm", "tran-type", "image-data", "id-code", "sfx", "photo", "gender", "date-issued", "id-no", "year-applied", "control-no", "id-type", "PVC-No", "age") VALUES -- RTC record (tran-type 1): keyed by batch-no + app-no (1001, '2026-01-15', '10:30:00', 1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr. Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 0, NULL, '', 1, NULL, 5001, '', NULL, 1, NULL, '', 0, 0, 1, 'PVC-000001', 0), -- OSCA/senior record (tran-type 2): keyed by control-no + id-no; batch-no/app-no are NULL (2001, '2026-01-16', '09:15:00', NULL, NULL, NULL, '1941-11-18', '', 'Maria Clara Santos', '', '456 Mabini St, Quezon City', '', 'Santos', 'Maria', 'Clara', '', '', '', '', '', NULL, 0, NULL, '', 2, NULL, 0, '', NULL, 2, '2026-01-16', 'OSCA-2026-0001', 2026, 9001, 1, 'PVC-000901', 84);
```

Notes on the sample above

- `trans-no` is the universal key — every row has one. It must be unique (OpenEdge PRIMARY index `id-idx`).
- RTC rows fill `batch-no`/`app-no`; OSCA rows fill `control-no`/`id-no` and leave `batch-no`/`app-no` as NULL.
- OpenEdge logical fields (`printed`) are `0/1`; the MySQL mirror stores them as `'no'/'yes'`.
- A copy of these two rows is in `docs/load-id-print.sql`, runnable against the local `iddb` via `sqlexp`.

Version: 1.4
Date: September 8, 2026

Purpose: Sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing. Reads `"id-print"` from OpenEdge and mirrors it to MySQL's `id_print` table.

Architecture Overview

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Progress OpenEdge SQL Server ■ ← Source of truth (remote, provided by client) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ODBC ■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■ ■ OpenEdge ODBC Driver ■ ← PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe ■ (DataDirect 8.0, 64-bit) ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■■■■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■ ■ C# Windows Service ■ ← OpenEdgeBridge.exe (this application) ■ (.NET 8, self-contained) ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ MySQL Connector ■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■ ■ MySQL Server (XAMPP) ■ ← Local database, runs on the same PC ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ JDBC (mysql-connector-j-8.0.33.jar) ■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■■ ■ Entrust Instant ID ■ ← Card issuance system ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Data Flow

OpenEdge → MySQL (One-Way Sync)

```
OpenEdge "id-print" ■■■ C# Service ■■■ MySQL id_print table (read) (sync loop) (upsert)
```

By default all 37 fields from the `id-print` table are synced, including:
- `trans-no`, `trans-date`, `trans-time`
- `batch-no`, `batch-date`, `app-no`, `dob`
- `cp-no`, `fullname`, `hcp-fullname`, `address`, `hcp-cpno`
- `lname`, `fname`, `mname`
- `spa-docno`, `page-no`, `book-no`, `series-no`, `spa-atty`, `spa-dt-issue`
- `printed`, `printed-dt`, `printed-tm`
- `tran-type`, `image-data`, `id-code`, `sfx`, `photo`
- `gender`, `date-issued`, `id-no`, `year-applied`, `control-no`
- `id-type`, `PVC-No`, `age`

The table holds **two ID kinds side by side**, distinguished by `tran-type`:
- `tran-type` **1 = RTC** → identified by `batch-no` + `app-no` (batched, many rows per run)
- `tran-type` **2 = OSCA/senior** → identified by `control-no` + `id-no` (single row, not batched)

`trans-no` is the **PRIMARY index** (`id-idx`) in OpenEdge and is present on **every** row of both kinds, so it is the universal key used by the bridge for fetching, deduplication, and the printed back-sync.

Client Filter — `Sync:Client` (RTC vs. OSCA)

A single deployment can be pointed at **one client** so it only handles that client's IDs. Set `Sync:Client` in `appsettings.json`, or override it with the `Sync__Client` environment variable:

Sync:Client	Rows fetched from OpenEdge	Columns projected	Rows written to MySQL
Both (default)	tran-type 1 **and** 2	all 37	both kinds
RTC	only tran-type = 1	RTC fields only	RTC only
OSCA	only tran-type = 2	OSCA fields only	OSCA only

When a single client is selected the bridge:

- adds WHERE "tran-type" = ? to the OpenEdge read, so the other kind is never fetched;
- projects **only that client's columns** (per the RTC/OSCA field lists above) — an OSCA read omits batch-no, batch-date, app-no, page-no, book-no, series-no, spa-docno, spa-atty, spa-dt-issue, hcp-cpno, hcp-fullname, id-code; an RTC read omits age, year-applied, date-issued, control-no, id-no;
- scopes the incremental watermark (MAX(trans_no)) **and** the printed back-sync to that tran-type, so a higher trans-no belonging to the other kind can never mask un-synced rows;
- skips (with a warning) any row whose tran-type doesn't match, so **only the configured client is ever created in MySQL**;
- writes **only the columns that belong to that client**, so a per-client id_print table needs no unrelated columns. The bridge verifies this once per run and logs an error (skipping the cycle) if the table and Sync:Client disagree.

Sync:Client accepts Both, RTC, OSCA (case-insensitive; also SENIOR/2 for OSCA, 1 for RTC). Blank or unrecognised values fall back to Both.

Runs every N seconds (configurable in appsettings.json). Uses **check-then-update** upserts (SELECT by trans_no, then UPDATE or INSERT) to avoid duplicates. print_count is only inserted on first sync; subsequent updates do NOT overwrite it (Instant ID manages print_count).

Full vs. Incremental Sync

The bridge has two read modes, decided automatically on every cycle:

Mode	Trigger	What it reads from OpenEdge
FULL	No last_full_sync_utc recorded yet (first run, or fresh MySQL)	All rows in pub."id-print" — seeds/refreshes the entire MySQL table
INCREMENTAL	A full sync has already been recorded	Only rows with trans-no greater than the highest trans_no already in MySQL

The incremental watermark is trans-no (not batch-no) because trans-no is the table-wide PRIMARY key that exists on every row — RTC batches and OSCA single records alike — whereas batch-no is NULL on OSCA/senior rows. New records always carry a newer trans-no, so incremental sync picks up both kinds.

OpenEdge id-print records are **write-once** — new data arrives only as new transactions/batches; existing rows are never edited. Therefore, with the default Sync:FullSyncIntervalMinutes = 0, the full sync runs **once** (the initial seed) and every later cycle is incremental. Restarting the service or rebooting the PC does **not** cause another full sync — the timestamp is persisted in MySQL's sync_state table (see Database Tables).

If `Sync:FullSyncIntervalMinutes` is set to a positive value (e.g., `60`), a full sync also runs when that many minutes have elapsed since the last successful full sync — useful only if OpenEdge rows ever start getting edited in place.

> **Note:** `signature_data`, `qr_data`, and `fingerprint_data` are **not** part of this sync — they are MySQL-only columns populated by Entrust Instant ID and never touched by the bridge (see Database Tables).

MySQL → OpenEdge (Printed Back-Sync)

```
MySQL id_print ■■■■■■■■ C# Service ■■■■■■■■ OpenEdge "id-print" (print_count > 0) (check + mark)
(SET printed = 1 + fields below)
```

When Instant ID increments `print_count > 0` on a record, the bridge detects it and, for the matching OpenEdge row (addressed by `trans-no` — not `app-no`, because OSCA/senior rows have no `app-no`), it writes back:

OpenEdge column	Written when
`"printed"`	always — set to `1`
`"printed-dt"`	MySQL `printed_dt` when the printer stamped it; otherwise the bridge stamps the current date
`"printed-tm"`	MySQL `printed_tm` when the printer stamped it; otherwise the bridge stamps the current time (`HH:mm:ss`)
`"photo"`	MySQL `photo` has content
`"image-data"`	MySQL `image_data` has content

`"printed-dt"` / `"printed-tm"` thus always end up populated: the **printer's own stamp wins** when MySQL has one, otherwise the bridge fills in its current date/time (OpenEdge's columns are always empty for these rows). `photo` / `image-data` are written only when they carry content, so they never blank out what OpenEdge already holds. `signature_data`, `qr_data` and `fingerprint_data` are MySQL-only and have no OpenEdge counterpart, so they are never pushed back.

> **`"PVC-No"` is never written back.** In MySQL `pvc_no` always holds a value, so pushing it would overwrite OpenEdge's `PVC-No` on every back-sync. It remains a strictly OpenEdge → MySQL field.

> This is the **only** MySQL → OpenEdge write in the bridge. All other columns flow one way, OpenEdge → MySQL.

Prerequisites

On the Target Windows PC

#	Component	Version / File	Notes
1	**Windows OS**	Windows 10/11 or Windows Server 2016+	64-bit required

2	**OpenEdge ODBC Driver**	`PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe`	Install, configure a System DSN
3	**XAMPP**	Latest (8.x)	Provides local MySQL server
4	**MySQL Connector/J**	`mysql-connector-j-8.0.33.jar`	For Entrust Instant ID to connect to MySQL
5	**OpenEdgeBridge**	Published `publish/` folder	Self-contained, no .NET runtime needed

---

Local Development — Installing Progress OpenEdge Database

> **Use this section when no remote OpenEdge server is available.**
> You can install Progress OpenEdge locally, create a database, and run the SQL below.

Step 0-A: Download & Install Progress OpenEdge

1. Go to https://www.progress.com/openedge/downloads
2. Download **Progress OpenEdge Developer Studio** or **Progress OpenEdge Personal Edition** (free for development)
3. Run the installer (typically `PROGRESS_OE_12.x_WIN_64.exe`) as Administrator
4. Choose **Typical** or **Complete** installation
5. Default install path: `C:\Progress\OpenEdge\`

After installation, a **"Proenv"** shortcut will appear in your Start Menu.
Proenv is a command prompt pre-configured with all OpenEdge environment variables.

Step 0-B: Create a Local OpenEdge Database

Open **Proenv** as Administrator and run:

```
:: Create a folder for your database mkdir C:\OpenEdgeDatabases cd C:\OpenEdgeDatabases :: Create an empty database called 'iddb' prodb iddb empty
```

This creates `iddb.db` and related files in `C:\OpenEdgeDatabases\`.

Step 0-C: Start the Database Server

In **Proenv**, run:

```
:: Start the database for multi-user access with both 4GL and SQL enabled proserve "C:\OpenEdgeDatabases\iddb.db" -S 11918 -ServerType Both -n 51 -L 8192
```

Parameter	Meaning

`-S 11918`	Service port (SQL and 4GL clients connect on this port)
`-n 51`	Maximum concurrent users
`-L 8192`	Lock table size (entries)
`-ServerType Both`	**Required** — enables both 4GL and SQL-92 engine

The database is now running. Keep this Proenv window open.

Step 0-D: Configure the ODBC System DSN

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab, click **Add...**
3. Select **Progress OpenEdge 12.x Driver** (or "DataDirect OpenEdge SQL Server")
4. Configure:
5. **Data Source Name:** `OpenEdgeInstantID`
6. **Description:** `OpenEdge Instant ID Database`
7. **Host Name:** `localhost`
8. **Port Number:** `11918`
9. **Database Name:** `iddb`
10. **User ID:** `admin`
11. Click **Test Connect** → should succeed
12. Click **OK**

Step 0-E: Create the Database Schema (`.df` File)

The table schema is defined in a `.df` (Data Definition) file. Create a file named `iddb.df` in `C:\OpenEdgeDatabases\` with the following content:

```
ADD TABLE "id-print" AREA "Schema Area" DUMP-NAME "id-print" ADD FIELD "batch-no" OF "id-print" AS
integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 2 MAX-WIDTH 4 ORDER 10 ADD FIELD "batch-date" OF
"id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 3 MAX-WIDTH 4 ORDER 20 ADD FIELD
"date-issued" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 29 MAX-WIDTH 4 ORDER
280 ADD FIELD "app-no" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 4
MAX-WIDTH 4 ORDER 30 ADD FIELD "dob" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION
5 MAX-WIDTH 4 ORDER 40 ADD FIELD "cp-no" OF "id-print" AS character FORMAT "x(15)" INITIAL ""
POSITION 6 MAX-WIDTH 30 ORDER 50 ADD FIELD "fullname" OF "id-print" AS character FORMAT "x(65)"
INITIAL "" POSITION 7 MAX-WIDTH 130 ORDER 60 ADD FIELD "hcp-fullname" OF "id-print" AS character
FORMAT "x(65)" INITIAL "" POSITION 8 MAX-WIDTH 130 ORDER 70 ADD FIELD "address" OF "id-print" AS
character FORMAT "x(65)" INITIAL "" POSITION 9 MAX-WIDTH 130 ORDER 80 ADD FIELD "hcp-cpno" OF
"id-print" AS character FORMAT "x(15)" INITIAL "" POSITION 10 MAX-WIDTH 30 ORDER 90 ADD FIELD
"lname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 11 MAX-WIDTH 110 ORDER 100
ADD FIELD "fname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 12 MAX-WIDTH 110
ORDER 110 ADD FIELD "mname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 13
MAX-WIDTH 110 ORDER 120 ADD FIELD "spa-docno" OF "id-print" AS character FORMAT "x(25)" INITIAL ""
POSITION 14 MAX-WIDTH 50 ORDER 130 ADD FIELD "page-no" OF "id-print" AS character FORMAT "x(20)"
INITIAL "" POSITION 15 MAX-WIDTH 40 ORDER 140 ADD FIELD "book-no" OF "id-print" AS character
```

```
FORMAT "x(20)" INITIAL "" POSITION 16 MAX-WIDTH 40 ORDER 150 ADD FIELD "series-no" OF "id-print"
AS character FORMAT "x(20)" INITIAL "" POSITION 17 MAX-WIDTH 40 ORDER 160 ADD FIELD "spa-atty" OF
"id-print" AS character FORMAT "x(65)" INITIAL "" POSITION 18 MAX-WIDTH 130 ORDER 170 ADD FIELD
"spa-dt-issue" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 19 MAX-WIDTH 4 ORDER
180 ADD FIELD "printed" OF "id-print" AS logical FORMAT "yes/no" INITIAL "no" POSITION 20
MAX-WIDTH 1 ORDER 190 ADD FIELD "printed-dt" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ?
POSITION 21 MAX-WIDTH 4 ORDER 200 ADD FIELD "printed-tm" OF "id-print" AS character FORMAT "x(35)"
INITIAL "" POSITION 22 MAX-WIDTH 70 ORDER 210 ADD FIELD "tran-type" OF "id-print" AS integer
DESCRIPTION "1 = for RTC, 2=senior" FORMAT "->,>>>,>>9" INITIAL "0" POSITION 23 MAX-WIDTH 4 ORDER
220 ADD FIELD "image-data" OF "id-print" AS raw FORMAT "x(55)" INITIAL "" POSITION 24 MAX-WIDTH
110 ORDER 230 ADD FIELD "id-code" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0"
POSITION 25 MAX-WIDTH 4 ORDER 240 ADD FIELD "sfx" OF "id-print" AS character FORMAT "x(15)"
INITIAL "" POSITION 26 MAX-WIDTH 30 ORDER 250 ADD FIELD "photo" OF "id-print" AS blob FORMAT
"x(8)" INITIAL ? POSITION 27 LOB-AREA "Schema Area" LOB-BYTES 104857600 LOB-SIZE 100M ORDER 260
ADD FIELD "gender" OF "id-print" AS integer DESCRIPTION "1=Female, 2=Male" FORMAT "->,>>>,>>9"
INITIAL "0" POSITION 28 MAX-WIDTH 4 ORDER 270 ADD FIELD "year-applied" OF "id-print" AS integer
FORMAT ">>>9999" INITIAL "0" POSITION 31 MAX-WIDTH 4 ORDER 300 ADD FIELD "control-no" OF
"id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 32 MAX-WIDTH 4 ORDER 310 ADD FIELD
"id-no" OF "id-print" AS character FORMAT "x(25)" INITIAL "" POSITION 33 MAX-WIDTH 50 ORDER 320
ADD FIELD "id-type" OF "id-print" AS integer DESCRIPTION "1=New, 2=ReIssue/Reprint" FORMAT
"->,>>>,>>9" INITIAL "0" POSITION 34 MAX-WIDTH 4 ORDER 330 ADD FIELD "PVC-No" OF "id-print" AS
character FORMAT "x(35)" INITIAL "" POSITION 35 MAX-WIDTH 70 ORDER 340 ADD FIELD "age" OF
"id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 36 MAX-WIDTH 4 ORDER 350 ADD FIELD
"trans-no" OF "id-print" AS int64 FORMAT "->,>>>,>>9" INITIAL "0" POSITION 37 MAX-WIDTH 8 ORDER
360 ADD FIELD "trans-date" OF "id-print" AS date FORMAT "99/99/99" INITIAL ? POSITION 38 MAX-WIDTH
4 ORDER 370 ADD FIELD "trans-time" OF "id-print" AS character FORMAT "x(30)" INITIAL "" POSITION
39 MAX-WIDTH 60 ORDER 380 ADD INDEX "id-idx" ON "id-print" AREA "Schema Area" PRIMARY INDEX-FIELD
"trans-no" ASCENDING ADD INDEX "oscaid-idx" ON "id-print" AREA "Schema Area" INDEX-FIELD
"control-no" ASCENDING INDEX-FIELD "id-no" ASCENDING INDEX-FIELD "tran-type" ASCENDING ADD INDEX
"rtcid-idx" ON "id-print" AREA "Schema Area" INDEX-FIELD "batch-no" ASCENDING INDEX-FIELD "app-no"
ASCENDING INDEX-FIELD "tran-type" ASCENDING . PSC cpstream=ISO8859-1 . 0000005403
```

Then, in **Proenv**, load the `.df` file into the database:

```
cd C:\OpenEdgeDatabases prodict iddb
```

In the Data Dictionary menu, select **Admin → Load Data Definitions → .df file**, then choose `iddb.df`.

> **Note:** The database name is `iddb` and the schema resides under the `PUB` area (default).

Database Tables

`id-print` / `id_print`

> Converted from the provided `.df` schema.
> **Paste each CREATE TABLE directly into your SQL editor and run it.**

OpenEdge SQL — CREATE TABLE

```
CREATE TABLE "id-print" ( "trans-no" BIGINT NOT NULL, "trans-date" DATE NULL, "trans-time"
VARCHAR(30) NOT NULL DEFAULT '', "batch-no" INTEGER NOT NULL DEFAULT 0, "batch-date" DATE NULL,
```

```
"app-no" INTEGER NOT NULL DEFAULT 0, "dob" DATE NULL, "cp-no" VARCHAR(15) NOT NULL DEFAULT '',
"fullname" VARCHAR(65) NOT NULL DEFAULT '', "hcp-fullname" VARCHAR(65) NOT NULL DEFAULT '',
"address" VARCHAR(100) NOT NULL DEFAULT '', "hcp-cpno" VARCHAR(15) NOT NULL DEFAULT '', "lname"
VARCHAR(55) NOT NULL DEFAULT '', "fname" VARCHAR(55) NOT NULL DEFAULT '', "mname" VARCHAR(55) NOT
NULL DEFAULT '', "spa-docno" VARCHAR(25) NOT NULL DEFAULT '', "page-no" VARCHAR(20) NOT NULL
DEFAULT '', "book-no" VARCHAR(20) NOT NULL DEFAULT '', "series-no" VARCHAR(20) NOT NULL DEFAULT
'', "spa-atty" VARCHAR(65) NOT NULL DEFAULT '', "spa-dt-issue" DATE NULL, "printed" BIT NOT NULL
DEFAULT 0, "printed-dt" DATE NULL, "printed-tm" VARCHAR(35) NOT NULL DEFAULT '', "tran-type"
INTEGER NOT NULL DEFAULT 0, "image-data" VARBINARY(110) NULL, "id-code" INTEGER NOT NULL DEFAULT
0, "sfx" VARCHAR(15) NOT NULL DEFAULT '', "photo" BLOB NULL, "gender" INTEGER NOT NULL DEFAULT 0,
"date-issued" DATE NULL, "id-no" VARCHAR(25) NOT NULL DEFAULT '', "year-applied" INTEGER NOT NULL
DEFAULT 0, "control-no" INTEGER NOT NULL DEFAULT 0, "id-type" INTEGER NOT NULL DEFAULT 0, "PVC-No"
VARCHAR(35) NOT NULL DEFAULT '', "age" INTEGER NOT NULL DEFAULT 0, PRIMARY KEY ("trans-no") );
```

Note: OpenEdge requires double-quotes around field names containing hyphens (`-`).
The schema resides in the `PUB` area (default).
`BIT` represents the OpenEdge `LOGICAL` type (yes/no → 1/0).
`VARBINARY(110)` stores the raw image binary data (MAX-WIDTH 110 from the schema).
`BLOB` stores the cardholder photo (LOB-SIZE 100M from the schema).

MySQL — CREATE TABLE (combined — `Sync:Client = Both`)

A single-client deployment can instead create a slimmer table containing only that client's columns — see **Per-Client Tables** below.

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, trans_no BIGINT NULL COMMENT
'Universal transaction no (OpenEdge PRIMARY key id-idx)', trans_date DATE NULL COMMENT
'Transaction date', trans_time VARCHAR(30) NULL COMMENT 'Transaction time', batch_no INT NULL,
batch_date DATE NULL, app_no INT NULL, dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65)
NULL, hcp_fullname VARCHAR(65) NULL, address VARCHAR(100) NULL, hcp_cpno VARCHAR(15) NULL, lname
VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, spa_docno VARCHAR(25) NULL,
page_no VARCHAR(20) NULL, book_no VARCHAR(20) NULL, series_no VARCHAR(20) NULL, spa_atty
VARCHAR(65) NULL, spa_dt_issue DATE NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches
OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL
COMMENT '1 = for RTC, 2 = senior', image_data LONGBLOB NULL, id_code INT NULL, sfx VARCHAR(15)
NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', date_issued DATE NULL,
id_no VARCHAR(25) NULL, year_applied INT NULL, control_no INT NULL, id_type INT NULL COMMENT '1 =
New, 2 = ReIssue/Reprint', pvc_no VARCHAR(35) NULL, age INT NULL, signature_data LONGBLOB NULL
COMMENT 'Signature image; written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)',
qr_data LONGBLOB NULL COMMENT 'QR code image; written by Entrust Instant ID (MySQL only, NOT
synced from OpenEdge)', fingerprint_data LONGBLOB NULL COMMENT 'Fingerprint image (OSCA/senior
only); written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)', print_count INT NULL
COMMENT 'Incremented by Instant ID; triggers printed=yes back-sync', PRIMARY KEY (id), UNIQUE
INDEX idx_trans_no (trans_no), INDEX idx_rtc_key (tran_type, batch_no, app_no), INDEX idx_osca_key
(tran_type, control_no, id_no), INDEX idx_print_sync (printed, print_count, trans_no) )
ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Note: `id` is an internal auto-increment primary key. `trans_no` is the **universal business key** — it mirrors OpenEdge's PRIMARY index (`id-idx`) and exists on **every** row, both RTC (`tran-type` 1) and OSCA/senior (`tran-type` 2) alike. The bridge uses it for deduplication (it SELECTs by `trans_no` first, then UPDATEs or INSERTs as needed). Unlike `batch_no`/`app_no` — which are NULL on OSCA rows — `trans_no` is never NULL, which is why it is the sync watermark too. `idx_rtc_key` and `idx_osca_key` are just lookup indexes for the type-specific business keys.

MySQL-only columns (`signature_data`, `qr_data`, `fingerprint_data`): These store the cardholder's signature, QR-code, and fingerprint images and are **MySQL only**. They do **not** exist in the OpenEdge `"id-print"` table and are **never read from or written back to OpenEdge**. Entrust Instant ID populates them directly in MySQL. The bridge deliberately leaves them untouched — it does not include them in its INSERT/UPDATE, so syncing from OpenEdge will never overwrite them. `signature_data` and `qr_data` are used by **both** RTC and OSCA; `fingerprint_data` is used by **OSCA/senior** only.

Already have the table? If you created `id_print` before these columns existed, add them with:

```
sql ALTER TABLE id_print MODIFY COLUMN address VARCHAR(100) NULL, CHANGE COLUMN id_type tran_type INT NULL COMMENT '1 = for RTC, 2 = senior', ADD COLUMN image_data LONGBLOB NULL AFTER tran_type, ADD COLUMN sfx VARCHAR(15) NULL AFTER id_code, ADD COLUMN photo LONGBLOB NULL AFTER sfx, ADD COLUMN gender INT NULL COMMENT '1 = Female, 2 = Male' AFTER photo, ADD COLUMN date_issued DATE NULL AFTER gender, MODIFY COLUMN id_no VARCHAR(25) NULL AFTER gender, ADD COLUMN year_applied INT NULL AFTER id_no, ADD COLUMN control_no INT NULL AFTER year_applied, ADD COLUMN id_type INT NULL COMMENT '1 = New, 2 = ReIssue/Reprint' AFTER control_no, ADD COLUMN pvc_no VARCHAR(35) NULL AFTER id_type, ADD COLUMN age INT NULL AFTER pvc_no, ADD COLUMN trans_no BIGINT NULL AFTER age, ADD COLUMN trans_date DATE NULL AFTER trans_no, ADD COLUMN trans_time VARCHAR(30) NULL AFTER trans_date, ADD COLUMN signature_data LONGBLOB NULL AFTER trans_time, ADD COLUMN qr_data LONGBLOB NULL AFTER signature_data, ADD COLUMN fingerprint_data LONGBLOB NULL AFTER qr_data, DROP INDEX idx_app_no, DROP INDEX idx_batch_no, ADD UNIQUE INDEX idx_trans_no (trans_no), ADD INDEX idx_rtc_key (tran_type, batch_no, app_no), ADD INDEX idx_osca_key (tran_type, control_no, id_no), ADD INDEX idx_print_sync (printed, print_count, trans_no);
```

Schema update (2026-09-08): The OpenEdge `.df` now has `trans-no` (int64) as its PRIMARY index (`id-idx`) and adds `id-type`, `PVC-No`, `age`, `trans-date`, `trans-time`. `trans-no` is the universal business key and incremental watermark (present on both RTC and OSCA rows, unlike `batch-no`/`app_no` which are NULL on OSCA rows). Reload the updated `.df` (Step 0-E), then run the ALTER above on MySQL.

MySQL — Per-Client Tables (`Sync:Client = RTC` or `OSCA`)

For a single-client deployment, create `id_print` with **only that client's columns** — no unrelated columns. The bridge writes only the columns that belong to the configured client, so a table built from either script below works. (The combined table above works too; its extra columns simply stay `NULL`.) If the table and `Sync:Client` disagree, the bridge logs an error naming the missing columns and skips the cycle — it checks once per run.

The Entrust-only image columns (`signature_data`, `qr_data`, and `fingerprint_data` for OSCA) are never written by the bridge; they exist only for Entrust Instant ID.

RTC only — `Sync:Client = "RTC"`

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, trans_no BIGINT NULL COMMENT 'Universal transaction no (OpenEdge PRIMARY key id-idx)', trans_date DATE NULL COMMENT 'Transaction date', trans_time VARCHAR(30) NULL COMMENT 'Transaction time', batch_no INT NULL, batch_date DATE NULL, app_no INT NULL, dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65) NULL, hcp_fullname VARCHAR(65) NULL, address VARCHAR(100) NULL, hcp_cpno VARCHAR(15) NULL, lname VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, spa_docno VARCHAR(25) NULL, page_no VARCHAR(20) NULL, book_no VARCHAR(20) NULL, series_no VARCHAR(20) NULL, spa_atty VARCHAR(65) NULL, spa_dt_issue DATE NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL COMMENT '1 = RTC', image_data LONGBLOB NULL, id_code INT NULL, sfx VARCHAR(15) NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', id_type INT NULL COMMENT '1 = New, 2 = ReIssue/Reprint', pvc_no VARCHAR(35) NULL, signature_data LONGBLOB NULL COMMENT 'Signature image; written by Entrust Instant ID (MySQL only)', qr_data LONGBLOB NULL COMMENT 'QR code image;
```

```
written by Entrust Instant ID (MySQL only)', print_count INT NULL COMMENT 'Incremented by Instant ID; triggers printed=yes back-sync', PRIMARY KEY (id), UNIQUE INDEX idx_trans_no (trans_no), INDEX idx_rtc_key (tran_type, batch_no, app_no), INDEX idx_print_sync (printed, print_count, trans_no) ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

OSCA / senior only — `Sync:Client = "OSCA"`

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, trans_no BIGINT NULL COMMENT 'Universal transaction no (OpenEdge PRIMARY key id-idx)', trans_date DATE NULL COMMENT 'Transaction date', trans_time VARCHAR(30) NULL COMMENT 'Transaction time', dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65) NULL, address VARCHAR(100) NULL, lname VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, sfx VARCHAR(15) NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL COMMENT '2 = OSCA/senior', image_data LONGBLOB NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', id_type INT NULL COMMENT '1 = New, 2 = ReIssue/Reprint', pvc_no VARCHAR(35) NULL, age INT NULL, control_no INT NULL, date_issued DATE NULL, id_no VARCHAR(25) NULL, year_applied INT NULL, signature_data LONGBLOB NULL COMMENT 'Signature image; written by Entrust Instant ID (MySQL only)', qr_data LONGBLOB NULL COMMENT 'QR code image; written by Entrust Instant ID (MySQL only)', fingerprint_data LONGBLOB NULL COMMENT 'Fingerprint image (OSCA/senior only); written by Entrust Instant ID (MySQL only)', print_count INT NULL COMMENT 'Incremented by Instant ID; triggers printed=yes back-sync', PRIMARY KEY (id), UNIQUE INDEX idx_trans_no (trans_no), INDEX idx_osca_key (tran_type, control_no, id_no), INDEX idx_print_sync (printed, print_count, trans_no) ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

> **Already have a combined table?** It keeps working with any `Sync:Client` value — the columns that don't belong to the selected client are simply never written (they stay `NULL`). No migration is required unless you want the slimmer schema.

`sync_state` (bridge state table — MySQL only, auto-created)

The bridge keeps its own bookkeeping in a tiny table called `sync_state`. **You do not need to create it manually** — the bridge runs `CREATE TABLE IF NOT EXISTS` on demand, so it appears in MySQL automatically after the first sync cycle.

```
CREATE TABLE IF NOT EXISTS sync_state ( state_key VARCHAR(64) NOT NULL PRIMARY KEY, state_value VARCHAR(64) NOT NULL, updated_utc DATETIME NOT NULL ) ENGINE=InnoDB;
```

Key	Meaning
`last_full_sync_utc`	UTC timestamp of the last successful FULL sync when `Sync:Client` is `Both`. Used to decide whether the next cycle can be INCREMENTAL. Survives restarts, so a reboot never forces a redundant full sync.
`last_full_sync_utc_rtc`	Same, for a deployment with `Sync:Client = RTC`.
`last_full_sync_utc_osca`	Same, for a deployment with `Sync:Client = OSCA`.

> The full-sync timestamp key is **per client**, so switching `Sync:Client` on an existing PC seeds a fresh FULL sync for the newly selected client (the bridge re-reads that client's rows and upserts them; separate keys mean one client's state never hides another's).

Column Comparison — Plain English

Think of each row as **one person's ID-card application folder**. The columns are the details written on that folder.

OpenEdge name	MySQL name	Plain English meaning
`trans-no`	`trans_no`	**Transaction number** — the table-wide PRIMARY key (`id-idx`). Every row has one, RTC and OSCA alike; this is the bridge's universal matchmaker
`trans-date`	`trans_date`	**Transaction date** — when the record was created
`trans-time`	`trans_time`	**Transaction time** — time the record was created
`batch-no`	`batch_no`	**Group number** — which batch/run of RTC applications this person came in with (NULL for OSCA/senior rows)
`batch-date`	`batch_date`	**Date of that group** — when their batch was made
`app-no`	`app_no`	**Application number** — the RTC application file number (part of the RTC key with `batch-no`; NULL on OSCA/senior rows)
`dob`	`dob`	**Date of birth**
`cp-no`	`cp_no`	**The person's reference number** (their own control/card number)
`fullname`	`fullname`	**Full name** — complete name all in one
`hcp-fullname`	`hcp_fullname`	**Doctor's name** — the healthcare provider attached to this person
`address`	`address`	**Home address**
`hcp-cpno`	`hcp_cpno`	**Doctor's number** — the healthcare provider's own reference number
`lname`	`lname`	**Last name** (surname)
`fname`	`fname`	**First name**
`mname`	`mname`	**Middle name**
`spa-docno`	`spa_docno`	**Permission letter number** — the number of the Special Power of Attorney (SPA) document
`spa-atty`	`spa_atty`	**Lawyer's name** — the attorney who signed that permission letter

`spa-dt-issue`	`spa_dt_issue`	**Date the permission letter was issued**
`page-no`	`page_no`	**Page number** — which page in the record book this person is on
`book-no`	`book_no`	**Book number** — which record book
`series-no`	`series_no`	**Series number** — the serial number of the ID
`tran-type`	`tran_type`	**Kind of ID / transaction type** — `1` = regular (RTC), `2` = senior citizen (OSCA). RTC rows are keyed by `batch-no`+`app-no`; OSCA rows by `control-no`+`id-no`
`image-data`	`image_data`	**The person's photo** (picture file)
`id-code`	`id_code`	**ID code** — the code number printed on the card
`sfx`	`sfx`	**Name suffix** — e.g. Jr., Sr., III
`photo`	`photo`	**Photo** — the cardholder's photograph (BLOB image)
`gender`	`gender`	**Gender** — `1` = Female, `2` = Male
`date-issued`	`date_issued`	**Date the ID card was issued**
`id-no`	`id_no`	**ID number** — character value from OpenEdge
`year-applied`	`year_applied`	**Year the application was filed** (e.g. `2026`)
`control-no`	`control_no`	**Control number** — application/control tracking number (the OSCA/senior ID key, with `id-no`)
`id-type`	`id_type`	**Issue kind** — `1` = New, `2` = ReIssue/Reprint
`pvc-no`	`pvc_no`	**PVC number** — the blank card/PVC stock used
`age`	`age`	**Age** — used on OSCA/senior rows
`printed`	`printed`	**Has it been printed yet?** — `yes` or `no`
`printed-dt`	`printed_dt`	**Date it was printed**
`printed-tm`	`printed_tm`	**Time it was printed**

—	`signature_data`	**Person's signature picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in
—	`qr_data`	**QR code picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in
—	`fingerprint_data`	**Fingerprint picture** — *MySQL only*. OSCA/senior only. Not in OpenEdge, not synced back. Entrust fills this in
—	`print_count`	**How many times it got printed** — *MySQL only*. Instant ID counts this; it's never overwritten by the bridge
—	`id`	**Row number** — MySQL's own internal counter. Not business data, ignore it

30-second summary: The bridge copies the person's details (name, birthdate, address, photo, etc.) **from OpenEdge → MySQL**, like photocopying the folder. `trans_no` is the **matchmaker** — it exists on every row (RTC and OSCA/senior alike) and makes sure updates land on the right row. With `Sync:Client` you can restrict a deployment to a single client (`RTC` or `OSCA`). The four columns that exist **only in MySQL** (`signature_data`, `qr_data`, `fingerprint_data`, `print_count`) are filled in by the **Entrust machine**, and the bridge never touches them.

Step-by-Step Setup (Remote Server)

> **If you already have a remote OpenEdge server**, follow these steps.
> **If no server is available**, go back to Local Development — Installing Progress OpenEdge Database first.

1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab
3. Click **Add...**
4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure:
6. **Data Source Name:** e.g., `OpenEdgeInstantID`
7. **Host:** (provided by your OpenEdge team, or `localhost` if local)
8. **Port:** (provided by your OpenEdge team, or `11918` if local)

9. **Database:** (provided by your OpenEdge team, or `iddb` if local)
10. **User ID / Password:** (provided by your OpenEdge team)
11. Click **Test Connect** to verify, then **OK**

2. Create the OpenEdge Tables

Connect to your OpenEdge database using any ODBC-compatible SQL client (DBeaver, HeidiSQL, etc.) and paste the SQL from the Database Tables section above:

- Run the **`"id-print"`** CREATE TABLE

2-B. Grant Table Permissions to the Login User

The bridge connects to OpenEdge using the **User ID** configured in the ODBC DSN (e.g., `admin`). If your login user is **not** `admin`, it needs explicit permissions on the table — otherwise the bridge fails with an access-denied error when it starts.

Run the following from a DBA account (e.g., `admin`), replacing `your_login_user` with the actual user you put in the DSN:

```
GRANT SELECT, INSERT, UPDATE, DELETE ON pub."id-print" TO your_login_user;
```

> **Note:** `SELECT` is required to read records for the OpenEdge → MySQL sync, and `UPDATE` is required for the printed back-sync (`printed = yes`). `INSERT`/`DELETE` are included for completeness. `pub` is the default schema area, and the double quotes around `"id-print"` are required because of the hyphen in the table name.

3. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings
3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

4. Create the MySQL Database & Tables

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE instantid_bridge;
```

Then paste and run the MySQL CREATE TABLE statements from the Database Tables section above:

- Run the **`id_print`** CREATE TABLE

5. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder
2. Configure Entrust to point to the local MySQL database:

3. **Host:** `localhost`
4. **Port:** `3306`
5. **Database:** `instantid_bridge`
6. **User:** `root` (or a dedicated user)
7. **Table:** `id_print`

6. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\`.

Edit `appsettings.json`

```
{ "ConnectionStrings": { "OpenEdge": "DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;",
"MySql": "Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_password;" }, "Sync":
{ "IntervalSeconds": 10, "FullSyncIntervalMinutes": 0, "Client": "Both" } }
```

Setting	Description
`OpenEdge`	ODBC DSN name, user, and password
`MySql`	MySQL server, database, user, and password
`Sync:IntervalSeconds`	How often the sync loop runs (default: 10s)
`Sync:FullSyncIntervalMinutes`	Minutes between periodic FULL syncs. `0` = disabled (recommended; OpenEdge rows are write-once, so only the first run seeds). Positive value (e.g., `60`) = re-read the whole table that often, catching in-place edits if they ever happen
`Sync:Client`	Which client this deployment syncs: `Both` (default), `RTC`, or `OSCA`. `RTC`/`OSCA` fetch and write only that kind's rows (see Client Filter). Also settable via the `Sync__Client` environment variable

Test the executable manually

Before registering the service, open PowerShell and run the published executable from its deployment folder:

```
Set-Location C:\OpenEdgeBridge .\OpenEdgeBridge.exe
```

The bridge runs in the foreground and writes startup, connection, and sync errors to the PowerShell window. Confirm that rows appear in `instantid_bridge.id_print`, then press **Ctrl+C** to stop it.

> Stop the manual test before starting the Windows service so both processes do not run at the same time.

7. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" ` -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" ` -DisplayName "OpenEdge Bridge Service" ` -Description "Syncs id-print data between OpenEdge and MySQL for Entrust Instant ID" ` -StartupType Automatic Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge # Status should be: Running
```

Service Management Commands

```
Start-Service OpenEdgeBridge # Start Stop-Service OpenEdgeBridge # Stop Restart-Service OpenEdgeBridge # Restart (after config changes) Get-Service OpenEdgeBridge # Check status # Uninstall Stop-Service OpenEdgeBridge sc delete OpenEdgeBridge
```

Logging & Troubleshooting

Log Locations

Source	Location
File logs	`C:\OpenEdgeBridge\Logs\` (dated log files)
Windows Event Viewer	*Windows Logs* → *Application*, Source: `OpenEdgeBridge`

Common Issues

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or credentials	Test the ODBC DSN in ODBC Administrator with **Test Connect**
"MySQL connection string not found"	`appsettings.json` not in same folder as `.exe`	Ensure `appsettings.json` is in `C:\OpenEdgeBridge\`
Duplicate rows in MySQL	Missing UNIQUE index on `trans_no`	Run the `ALTER TABLE ... ADD UNIQUE INDEX idx_trans_no (trans_no)` SQL above
OpenEdge ODBC driver not found	Wrong bitness (32 vs 64)	Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64

`Failure getting schema read lock (12138)`	SQL engine not initialized for database	1. Verify `-ServerType Both` or `-ServerType SQL` on proserve 2. In sqlexp run: `UPDATE TABLE STATISTICS AND INDEX STATISTICS AND ALL COLUMN STATISTICS;` 3. If still failing, restart DB: `proshut` then `proserve` with `-SQLTrc 3`
MySQL not accessible from Entrust	MySQL not running or wrong port	Check XAMPP Control Panel; MySQL must be **Started**
OpenEdge SQL CREATE TABLE fails with "Invalid table name"	Hyphens in table/field names not quoted	Wrap names containing hyphens in double quotes: `"id-print"`, `"batch-no"`
`image-data` field too small	VARBINARY length insufficient	The OpenEdge schema specifies MAX-WIDTH 110; adjust as needed
Full sync on every start (log says `[FULL]`)	`sync_state` missing or MySQL unreachable at startup	The bridge auto-creates `sync_state`; after the first successful full sync it records a timestamp and later starts use `[INCREMENTAL]`. If MySQL is down at startup it falls back to FULL — that is safe and self-corrects once MySQL is reachable

---

Sync Interval Tuning

`IntervalSeconds` — how often the bridge checks for new batches

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": { "IntervalSeconds": 30 }
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

The interval is an **idle pause between cycles** (work → wait → work), not a fixed wall-clock schedule. A slow sync simply delays the next check; cycles never overlap.

`FullSyncIntervalMinutes` — how often the whole table is re-read

```
"Sync": { "FullSyncIntervalMinutes": 0 }
```

Value	Behavior

`0` (default, recommended)	FULL sync only on the **first run** (seeds MySQL). All later cycles are INCREMENTAL (`trans-no` > highest in MySQL). Restarts never re-trigger a full sync.
Positive (e.g., `60`)	Additionally re-reads the entire table when that many minutes have passed since the last full sync. Only needed if OpenEdge rows can be edited in place.

How the decision works: after every successful FULL sync the bridge writes the completion time to `sync_state.last_full_sync_utc` in MySQL. On each cycle it compares that timestamp against `FullSyncIntervalMinutes`. If MySQL is unreachable at startup, the bridge falls back to FULL — the safe default that guarantees a complete copy.

Why `0` is right for this deployment: OpenEdge `id-print` data is transaction-based and write-once — new records always carry a newer `trans-no` (RTC rows arrive in batches, OSCA/senior rows one at a time), and existing rows are not edited. The incremental query (trans-no greater than the current max in MySQL) therefore sees every new record of both kinds, and a periodic full re-read only wastes database load on a table that never changes. If the provider ever starts correcting existing rows, set this to `60` (hourly refresh) or similar.

`Client` — which client this deployment syncs

```
"Sync": { "Client": "Both" }
```

Value	Behavior
`Both` (default)	Syncs both RTC (`tran-type` 1) and OSCA/senior (`tran-type` 2) rows.
`RTC`	Fetches only `tran-type` = 1 rows, projects RTC columns only, and writes RTC rows only.
`OSCA`	Fetches only `tran-type` = 2 rows, projects OSCA columns only, and writes OSCA rows only.

Set this to the client this PC is deployed for. It can also be supplied through the environment as `Sync__Client` (double underscore instead of the colon). Blank or unrecognised values fall back to `Both`. See Client Filter for details.

Port Summary

Component	Port	Direction
OpenEdge SQL Server	Provided by OpenEdge team	Outbound
MySQL (XAMPP)	3306	Local
Entrust → MySQL	3306	Local via JDBC

No inbound ports are opened by the bridge itself.

Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

> **After upgrading from v1.1 or earlier:** the new `sync_state` table does not exist yet, so the first run performs one FULL sync to seed MySQL and record the timestamp. From then on, only new batches are synced — including across restarts.

Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)
- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database created in MySQL
- [ ] `id-print` / `id_print` table created in **both** OpenEdge and MySQL
- [ ] `sync_state` table present in MySQL (auto-created by the bridge after the first run)
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors